

ERIC WARMENHOVEN

Washington, DC area · eric@warmenhoven.org · [linkedin.com/in/ericwarmenhoven](https://www.linkedin.com/in/ericwarmenhoven)

PROFESSIONAL SUMMARY

Staff software engineer with 24+ years of experience building systems across the full stack, from embedded devices to cloud services. Proven track record of technical leadership on high-impact projects at Netflix, spanning streaming infrastructure on millions of consumer devices, engineering management, and cloud-based gaming platforms. Deep expertise in embedded systems, cross-platform SDKs, and constrained device optimization. Active open-source contributor with experience shipping software used by millions of users worldwide.

TECHNICAL SKILLS

LANGUAGES	C, C++, Objective-C, Swift, JavaScript, TypeScript, Java, Python, Perl, Shell scripting
TECHNOLOGIES	iOS/tvOS, Embedded Linux, WebAssembly, GraphQL, Node.js, WebKit, Docker, REST APIs, DRM systems
DOMAINS	Embedded Systems, Streaming Media, Gaming SDKs, Cross-platform Development, Distributed Systems, High Availability
TOOLS	Git, Perforce, Xcode, GDB, Make/CMake, CI/CD pipelines, Atlassian suite

PROFESSIONAL EXPERIENCE

Staff Software Engineer, Games SDK

February 2021 – Present

Netflix · Remote (Maryland)

- Lead iOS engineer on the Netflix Games SDK, enabling third-party game developers to integrate with Netflix's cloud-based Games Development Platform
- Drive technical leadership across feature design, architectural decisions, API design, and security reviews for a multi-language SDK (C++, Objective-C, Swift, C#, TypeScript)
- Design and implement network/transport layer architecture including GraphQL schema design and secure communication protocols
- Collaborate with game development partners to adapt the SDK to diverse requirements across platforms including iOS and Unity
- Contribute to WebAssembly-based approaches for TV gaming experiences using cloud technologies

Senior Software Engineer, Platform

September 2017 – February 2021

Netflix · Remote (Maryland)

- Owned Backend-For-Frontend (BFF) servers powering playback and UI on iOS and tvOS streaming applications
- Built and maintained Objective-C client libraries consuming BFF services, ensuring reliable video playback for millions of users
- Architected server infrastructure using Node.js (Restify) running in Docker containers with focus on scalability and reliability
- Partnered with device and server teams to define protocols and optimize client-server communication patterns

Engineering Manager, TV Platform

January 2013 – August 2017

Netflix · San Francisco Bay Area

- Led engineering team responsible for video player on Smart TVs and TV-connected devices reaching hundreds of millions of subscribers
- Owned JavaScript APIs enabling UI teams to control video playback across diverse device ecosystems
- Collaborated with server teams on protocols for fetching movie metadata and DRM license acquisition
- Managed custom software stack balancing flexible, updatable JavaScript with performance-critical C++ components
- Drove technical strategy for embedded device platform spanning multiple hardware generations

Senior Software Engineer, Core SDK

March 2010 – January 2013

Netflix · Los Gatos, CA

- Tech lead for Netflix's core C++ SDK powering streaming on consumer electronics devices worldwide
- Ported WebKit to Nintendo Wii, establishing UI framework for Netflix's first gaming console deployment
- Designed and maintained JavaScript APIs for custom adaptive streaming player across multiple platforms
- Advised cross-functional teams on SDK adaptations for PS3, Wii, Android, iOS, and ChromeOS
- Contributed to SDK architecture supporting both UI framework and adaptive bitrate streaming player

Member of Technical Staff, Software Engineer

March 2007 – February 2010

TiVo · Alviso, CA

- Delivered TiVo user experience to OCAP-compliant cable set-top boxes, expanding platform reach to cable providers
- Developed Java-based data management and user interface components for embedded cable box platform
- Implemented low-level C components for OCAP middleware stack including JVM optimizations
- Designed and built automated test infrastructure for rapid regression detection
- Optimized memory usage and file system performance through system-level service redesign

Staff Software Engineer

June 2002 – March 2007

Brocade Communications · San Jose, CA

- Ported Fibre Channel Router from Rhapsody (NetBSD-based) to Brocade's Linux platform
- Designed high-availability architecture for multiple subsystems including the FC router
- Maintained Reliable Commit Service ensuring consistency of changes across distributed systems
- Refactored build infrastructure reducing build times from hours to minutes
- Awarded three patents for innovations in Fibre Channel networking technology

OPEN SOURCE CONTRIBUTIONS

Libretro

2021 – Present

Contributor

Active contributor to RetroArch cross-platform gaming emulation frontend and many libretro cores, focusing on macOS/iOS/tvOS platform support, CI/CD pipeline using gitlab and docker, crash analysis, performance optimization, and build system improvements.

Debian

2005 – 2014

Developer

Maintain packages for the Debian Linux distribution.

Gaim/Pidgin

2000 – 2002

Lead Developer

Led development of multi-protocol instant messaging client. Redesigned networking for non-blocking I/O, created plugin architecture, and implemented support for AIM, Yahoo, MSN, and other protocols. Software now used by millions worldwide.

EDUCATION

Bachelor of Science in Computer Science

Graduated June 2002

University of California, Santa Barbara

PATENTS

Three patents awarded for Fibre Channel networking innovations at Brocade Communications (2002-2007)